

Data Management Plan

Detecting AI-Generated Text Using ML and NLP Techniques

GitHub: <https://github.com/SabirShah1402/Detecting-AI-Generated-Text-Project>

3.1 Overview of Datasets

Two publicly available, labelled datasets are used. Together they provide tens of thousands of real-world human and AI-generated text samples across five specialist domains and general writing, enabling robust training and cross-domain generalisation testing.

Attribute	HC3 Dataset	Kaggle Baseline Dataset
Source URL	https://huggingface.co/datasets/Hello-SimpleAI/HC3	https://www.kaggle.com/code/rsuhara/ai-generated-text-detection-quick-baseline
Provider	Hello-SimpleAI (HuggingFace Hub)	rsuhara (Kaggle Notebooks)
File Format	HuggingFace Datasets / Parquet	CSV via Kaggle notebook
Dataset Size	24,300 rows 147 MB raw	~487,000 rows ~1 GB
Key Columns	id, question, human_answers (list), chatgpt_answers (list), source	text (string), generated (int 0/1)
Domains	Reddit ELI5, Finance, Medicine, Open QA, Wiki CSAI	General / mixed writing styles, multiple LLMs
Label Schema	0 = human_answers, 1 = chatgpt_answers (implicit by list type)	0 = Human, 1 = AI-Generated (explicit column)
Language	English (+ Chinese subset, unused)	English
Licence	CC-BY-SA 4.0	Public research / Kaggle ToS
Paper / Citation	Guo et al., arXiv:2301.07597 (2023)	rsuhara, Kaggle 2023
Downloaded Size	147 MB	Varies by split used

3.2 Data Collection

- **HC3 Programmatic:** Loaded directly in Python via the HuggingFace Datasets library:
`load_dataset("Hello-SimpleAI/HC3", "all", split="train")`.
No manual download needed. The dataset commit hash is stored in the README to pin the version for reproducibility.
- **Kaggle Notebook-based:** Access the dataset and notebook at <https://www.kaggle.com/code/rsuhara/ai-generated-text-detection-quick-baseline>.
The CSV is downloaded via the Kaggle API (kaggle kernels output rsuhara/ai-generated-text-detection-quick-baseline -p ./data/raw/) or manually. A SHA-256 checksum is stored in the README to verify file integrity across team members.

3.3 Metadata

Every row in the merged dataset carries the following fields:

text (str — raw text), label (int: 0 or 1), source (str — HC3 domain or "kaggle"), dataset (str — "HC3" or "Kaggle"), word_count (int), char_count (int), avg_sentence_length (float), punctuation_density (float).

Schema is fully documented in /docs/metadata_schema.md in the repository.

3.4 Document Control — GitHub

Repository: <https://github.com/SabirShah1402/Detecting-AI-Generated-Text-Project>

All code, notebooks, and planning documents are version-controlled with Git. Branch strategy:

- main — protected; stable, reviewed code only. Merges require pull request + one peer approval minimum.
- dev — active integration branch. All feature work merges here first via pull request.
- feature/[task-id] — short-lived task branches (e.g., feature/T5-hybrid-cnn). Deleted after merge.

- Week 3 repo folder contains final versions of all six planning documents as required by the brief.

File naming conventions applied consistently across all project artefacts:

3.5 README File

README.md at the repository root documents: project title, description, and team members; dataset download instructions with source URLs and checksums; Python environment setup (pip install -r requirements.txt); notebook execution order (01 through 03 in sequence); model checkpoint locations and Google Drive download links; and Streamlit app launch instructions (streamlit run app.py). The README is updated at each milestone merge.

3.6 Security and Storage

- **Raw data:** Stored in /data/raw/ which is gitignored. Preprocessed splits (train.csv, val.csv, test.csv) are committed — they are clean, reproducible, and contain no sensitive information.
- **Model weights:** Stored in /models/ (gitignored due to file size). Shared via Google Drive link in README. Final model tagged as a GitHub Release at v1.0.
- **Access control:** Repository is private during development; made public at submission for examiner access. No API keys, credentials, or raw dataset files are committed. All secrets managed via a gitignored .env file.
- **Backup:** Each team member maintains a full local clone. Google Drive serves as secondary backup for large artefacts (raw datasets, model checkpoints, draft report versions).

3.7 Ethical Statement

Q1: Does the data come under GDPR requirements?

Neither dataset contains personally identifiable information as defined by GDPR (Regulation EU 2016/679). HC3 contains anonymised public forum responses (Reddit, Wikipedia) and programmatically generated AI answers, with no individual identifiers. The Kaggle dataset contains general writing samples with no personal data. No data subjects can be identified from either dataset.

Q2: Does the project conform to UH ethical policies?

Yes. The project uses pre-existing, publicly available secondary datasets for academic, non-commercial research. No human participants are recruited, no primary data is collected, and no surveys or experiments involving people are conducted. The research goal — detecting AI-generated content to support academic integrity and combat misinformation — is socially beneficial. The project falls within the standard low-risk academic research exemption under the University of Hertfordshire Research Ethics Policy.

Q3: Do you have permission to use the data?

Yes. HC3 is released under CC-BY-SA 4.0, which explicitly permits research and educational use with attribution (Guo et al., 2023, arXiv:2301.07597). The Kaggle notebook and associated dataset are published for public research use under Kaggle's Terms of Service, which permit non-commercial academic research. Both licences are compatible with this project.

Q4: Was the data collected ethically?

Yes. HC3 was created by Hello-SimpleAI researchers who sourced human answers from public platforms under their existing terms of service, and generated ChatGPT responses via OpenAI's API in compliance with OpenAI usage policies. The research was published transparently alongside a peer-reviewed paper (arXiv:2301.07597). The Kaggle notebook uses publicly accessible text samples compiled specifically for AI detection research. Both datasets were created and shared openly and ethically.